

- b. From the following tables find $f(x)$ and hence $f(4)$ using Newton's formula.

x	-1	1	2	3
y	-21	15	12	3

30. a. A rod is rotating in a plane. The following table gives the angle θ (in radians) through which the rod has turned for various values of time t (seconds). Calculate the angular velocity and angular acceleration of the rod at $t = 0.6$ seconds.

t	0	0.2	0.4	0.6	0.8	1.0
θ	0	0.12	0.49	1.12	2.02	3.20

(OR)

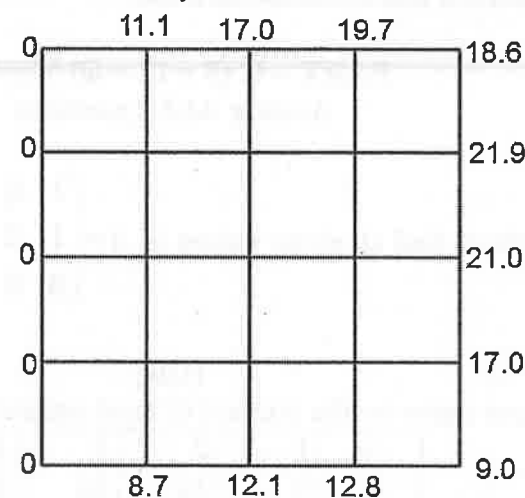
- b. Evaluate $\int_4^{5.2} \log_e x dx$ using Trapezoidal and Simpson's rules.

31. a. Using R.K method of fourth order compute $y(0.1), y(0.2)$ given $\frac{dy}{dx} + y + xy^2 = 0, y(0) = 1$, correct to 4 decimals.

(OR)

- b. Find $y(2)$ if $y(x)$ is the solution of $\frac{dy}{dx} = \frac{1}{2}(x + y)$ given $y(0) = 2, y(0.5) = 2.636, y(1) = 3.595, y(1.5) = 4.968$.

32. a. Find the Liebmann's method the values at the interior lattice points of the square region of the harmonic function u whose boundary values are as shown below.



(OR)

- b. Given $f_{xx} = f_t, f(0, t) = f(5, t) = 0, f(x, 0) = x^2(25 - x^2)$, find f in the range taking $h = 1$ and upto 5 seconds.

B.Tech. DEGREE EXAMINATION, NOVEMBER 2018
3rd to 7th Semester

15MA206 – NUMERICAL METHODS

(For the candidates admitted during the academic year 2015 – 2016 to 2017-2018)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 45 minutes and OMR sheet should be handed over to hall invigilator at the end of 45th minute.
(ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: Three Hours

Max. Marks: 100

PART – A (20 × 1 = 20 Marks)

Answer **ALL** Questions

- The principle of method of least squares state that the best fitting curve to the set of points is that for which the sum of the squares of residuals is
(A) Maximum (B) Minimum
(C) Greater than zero (D) Less than zero
- Newton-Raphson method is also known as the method of _____.
(A) Secant (B) Co-secant
(C) Tangents (D) Squares
- The convergence in the Gauss-Seidel method is roughly _____ as faster as the in Jacobi's method.
(A) Three times (B) Five times
(C) Two times (D) Ten times
- In Gauss-Elimination method for solving a system of linear algebraic equations, the triangularization leads to
(A) Lower triangular matrix (B) Upper triangular matrix
(C) Diagonal matrix (D) Singular matrix
- The $(n+1)^{\text{th}}$ difference of the n^{th} degree polynomial is
(A) Constant (B) Zero
(C) Positive (D) Negative
- $E \nabla =$
(A) E (B) 1
(C) ∇ (D) Δ
- The dependent value y , in the function $y = f(x)$ is known as
(A) Argument (B) Entry
(C) Constant (D) Variable
- $(1 + \Delta)(1 - \nabla) = ?$
(A) 0 (B) 1
(C) δ^2 (D) Δ

9. Simpson's three-eighth rule can be applied only when n is
 (A) Odd (B) Even
 (C) Prime (D) Multiple of 3

10. The error in the Trapezoidal is of order

- (A) $\frac{h^2}{12}$ (B) $\frac{h^3}{6}$
 (C) $\frac{3h^2}{8}$ (D) h^2

11. Numerical differentiation is the process of computing the value of

- (A) Δy_0 (B) ∇y_0
 (C) $\frac{\Delta y}{\Delta x}$ (D) $\frac{dy}{dx}$

12. The truncation error in Simpson's one third rule is

- (A) $|E| < \frac{(b-a)h^2}{12} M$ (B) $|E| < \frac{(b-a)h^4}{180} M$
 (C) $|E| < \frac{(b-a)h^4}{80} M$ (D) $|E| < \frac{(b-a)h^6}{840} M$

13. Taylor's series method is

- (A) Single step method (B) Multi step method
 (C) Iterative method (D) Trial and error method

14. How many prior values are required to predict the next value in Adam's methods

- (A) 1 (B) 2
 (C) 3 (D) 4

15. The error term in Milne's predictor formula is

- (A) $\frac{-19h}{720} \Delta^4 y_0'$ (B) $\frac{14h}{90} \Delta^4 y_0'$
 (C) $\frac{-h}{90} \Delta^4 y_0'$ (D) $\frac{14h}{45} \Delta^4 y_0'$

16. Given $y' = x + y$, $y(0) = 1$, then $y(0.1)$ by Euler's method is

- (A) 0.1 (B) 0.2
 (C) 2.1 (D) 1.1

17. The nature of the PDF $f_{xx} - 2f_{xy} = 0$ is

- (A) Laplace equation (B) Elliptic
 (C) Parabolic (D) Hyperbolic

18. Bender-Schmidt scheme converges for

- (A) $\lambda = 1$ (B) $\lambda = 1/2$
 (C) $\lambda = 3/2$ (D) $\lambda = 3$

19. The error in the diagonal formula is _____ times the error in the standard formula.

- (A) 1 (B) 2
 (C) 3 (D) 4

20. The equation $u_{xx} + u_{yy} = f(x, y)$ is also called

- (A) Laplace equation (B) Poisson's equation
 (C) One dimensional heat equation (D) Two dimensional heat equation

PART - B (5 × 4 = 20 Marks)

Answer ANY FIVE Questions

21. Find the real positive root of $3x - \cos x - 1 = 0$ by the method of tangents.

22. Express $f(x) = x^3 - 3x^2 + 5x + 7$ in terms of factorial polynomial taking $h = 2$ and find its differences.

23. Find the 7th term of the sequence 1, 4, 10, 20, 35, 56, ...

24. Evaluate $\int_{-3}^3 x^4 dx$, by using Trapezoidal rule with $h = 1$.

25. Solve $\frac{dy}{dx} = y^2 + 1$, given $y(0) = 0$ and get $y(0.2)$.

26. Classify the following PDE $(x+1)u_{xx} - 2(x+2)u_{xy} + (x+3)u_{yy} = 0$.

27. State Adam's predictor and corrector formula.

PART - C (5 × 12 = 60 Marks)

Answer ALL Questions

28. a. Using power method, find all eigen values of $A = \begin{pmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$.

(OR)

- b. Fit a second degree curve by the method of least squares, to the following data.

x	1	2	3	4	5
y	5	12	26	60	97

Also find y at $x = 6$.

29. a. From the data given below, find the number of students whose weight is between 60 and 70lbs.

Weight in lbs	0-40	40-60	60-80	80-100	100-120
No of students	250	120	100	70	50

(OR)